ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B1 · WATER CYCLE

The water cycle

★☆☆
Grade 5

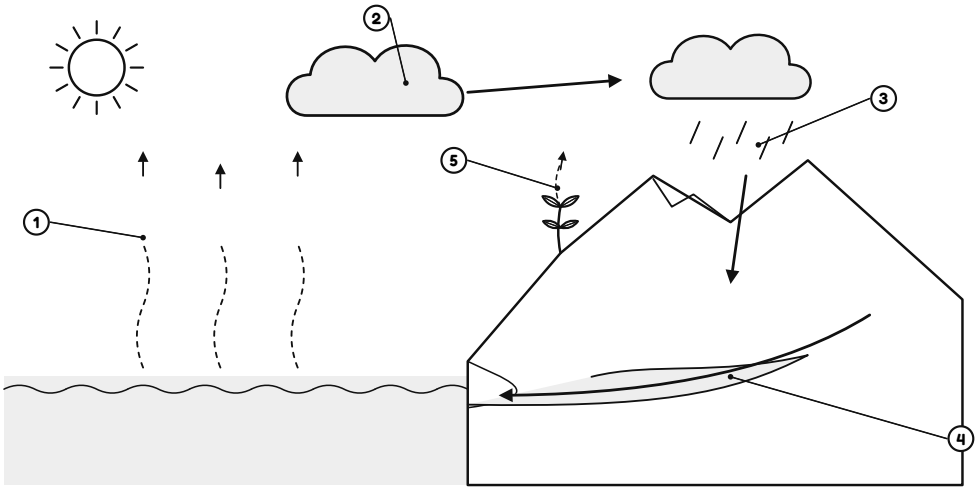
Learning goals

- Label the water cycle.
- Name the changes of state.

1 Label the water cycle. [5]

LABELS

evaporation **condensation** **collection** **precipitation** **transpiration**



(1) evaporation (2) condensation (3) precipitation

(4) collection (5) transpiration

2 Complete the table. [8]

Change	From	To	Caused by
evaporation	liquid	gas	heating
condensation	gas	liquid	cooling
melting	solid	liquid	heating
freezing	liquid	solid	cooling

3 Fill in the gaps. [4]

WORD BANK

Sun **100** **50** **condenses** **wind** **0**

Water boils at 100 °C and ice melts at 0 °C. Water vapour cools and condenses to make clouds. The Sun gives the energy that drives the water cycle.

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Rain water is salty because it comes from the sea.		✔	Rain is fresh; the salt stays behind when sea water evaporates.
2. Plants put water vapour into the air.	✔		
3. Most of Earth's water is fresh water.		✔	About 97% is salty sea water.

ANSWER KEY

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THINK IT THROUGH · B1 · WATER CYCLE

Explaining the cycle

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Grade 5

Learning goals

- Order and explain the water cycle.
- Link the cycle to everyday events.

1 **Number the stages to follow one drop of water from the Bay of Bengal back to the sea.** [3]

1	The Sun heats the sea and the water evaporates.
6	The river flows back into the sea.
3	It condenses into droplets and forms clouds.
5	Rain runs into a river.
4	Droplets join and fall as monsoon rain.
2	Water vapour rises and cools.

2 **Explain each observation using the words in brackets.** [6]

a) A cold glass of water gets wet on the outside. (*water vapour, condenses*)

Water vapour in the air touches the cold glass, cools and condenses into liquid droplets.

b) Puddles disappear on a sunny day. (*evaporates*)

The Sun heats the water and it evaporates into water vapour in the air.

c) Clothes dry faster on a windy day.

Wind carries away the water vapour, so more water can evaporate quickly.

3 **Why is the water cycle important for farmers in Bangladesh?** [2]

It brings fresh rain for crops like rice; it fills rivers and groundwater used for irrigation and drinking.

4 **How does pumping too much groundwater affect the water cycle in a city? Suggest one solution.** [2]

Water is taken out faster than rain can soak back in, so the water level falls. Solutions: rainwater harvesting, using river water, leaving green ground for rain to soak in.

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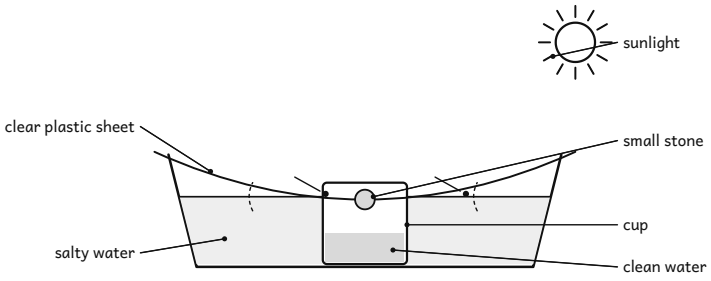
INVESTIGATE · B1 · WATER CYCLE

Make a solar still

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Learning goals

- Build a model that uses evaporation and condensation.
- Test a variable fairly.
- Use results to explain desalination.



You will need

a large bowl, a cup, salty water, clear plastic sheet, a small stone, sticky tape, a measuring cylinder

Stay safe

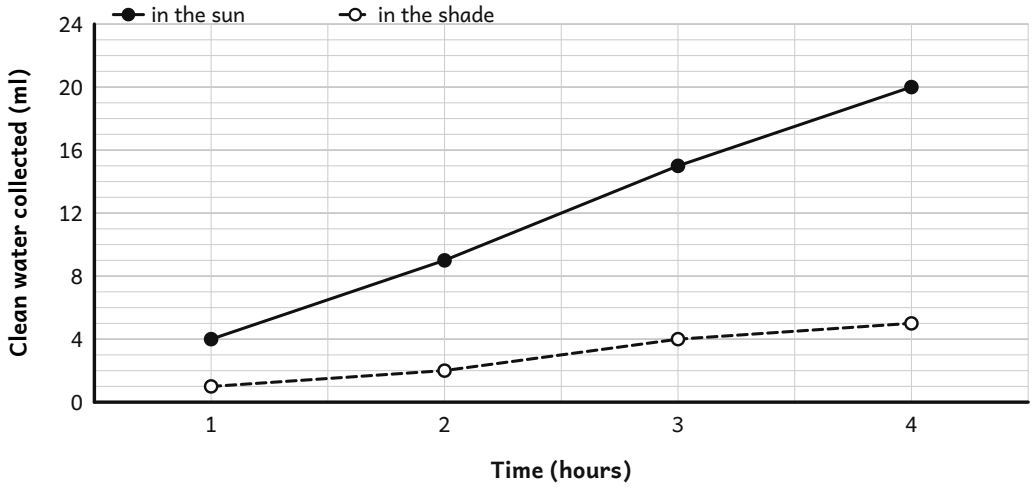
Do not taste anything in a science lab. Test for salt by letting one drop dry on a dark tile.

1 **How does the solar still work? Use the words *evaporates*, *condenses* and *salt*.** [3]

The Sun heats the salty water so water evaporates; the vapour touches the cool plastic and condenses into droplets; they run to the low point and drip into the cup. The salt is left behind in the bowl.

2 **Class 5B set up two identical stills, one in sunshine and one in shade, and measured the clean water every hour.** [4]

Time (hours)	1	2	3	4
In the sun (ml)	4	9	15	20
In the shade (ml)	1	2	4	5



3 **What is the variable they changed? Name two variables they kept the same.** [2]

Changed: sunshine or shade (amount of heat). Same: amount and saltiness of water, size of bowl and cup, type of plastic, time.

4 **Write a conclusion and explain it.** [2]

The still in the sun made much more clean water: more heat makes water evaporate faster, so more condenses on the plastic.

5 **Was the water in the cup salty? How could you check without tasting it?** [1]

No: the salt does not evaporate. Let a drop dry on a dark tile; no white salt crystals are left.

ANSWER KEY

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B1 · WATER CYCLE

Too much water, not enough to drink

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Grade 5

Learning goals

- Explain a real water problem using science.
- Compare solutions using evidence.
- Plan actions to protect water.

Science | Geography | Maths | Health | Climate change

In Satkhira and Khulna, in south-west Bangladesh, people are surrounded by water: rivers, shrimp ponds and the sea. But most of it is **too salty to drink**. Storm surges from cyclones such as Aila (2009) flooded ponds with sea water, and rising sea levels push salt further inland each year.

Many women and girls walk several kilometres every day to fetch drinking water. Some families collect **rainwater** from their roofs in big tanks during the monsoon. Villages also use **pond sand filters**, which clean water from a protected pond, and small **desalination plants** that use reverse osmosis.

Drinking salty water for a long time is bad for health, especially for pregnant women, whose blood pressure can rise.

Based on reports about coastal salinity in Bangladesh (WHO, UNICEF, Bangladesh research studies).

1 **Why is there a water problem in a place that has so much water?** [2]

Most of the water is salty (sea water, salty rivers and ponds); fresh drinking water is scarce.

2 **A family of 5 needs about 3 litres of drinking water per person each day.** [3]

a) How much drinking water does the family need in one day?

5 × 3 = 15 litres

b) How much in 30 days?

15 × 30 = 450 litres

c) Their rainwater tank holds 2000 litres. For about how many days will a full tank last?

2000 ÷ 15 ≈ 133 days (about 4 months)

3 **Compare three solutions.** [6]

Solution	One advantage	One disadvantage
rainwater tank	free, clean water; uses the monsoon rain	runs out in the dry season; tank costs money
pond sand filter	shares clean water with many families	the pond must be protected from salt and dirt
desalination plant	makes fresh water from salty water all year	expensive; needs electricity and repair

4 **Fetching water often stops girls from going to school. Why is safe water a fairness issue too?** [2]

Girls lose learning time and future chances; everyone has a right to safe water, so a nearby supply gives girls the same chance to learn.

ANSWER KEY

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REVIEW AND CHALLENGE · B1 · WATER CYCLE

Show what you know

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Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

W	N	F	C	D	T	V	A	U	R	M
E	U	F	T	O	P	V	E	E	R	O
R	V	S	N	C	N	N	G	P	A	N
S	B	A	O	O	L	D	M	C	I	S
S	U	B	P	D	V	O	E	J	N	O
S	R	N	C	O	T	A	U	N	G	O
A	E	V	C	U	R	R	P	D	S	N
L	R	G	O	L	H	A	I	O	P	E
T	H	C	W	C	P	S	T	V	U	G
F	M	G	S	H	D	F	S	E	E	R
C	P	P	R	A	A	G	A	K	J	R

cloud **condense** **evaporate** **monsoon** **rain** **river** **salt** **sun** **vapour**

2 **Which process forms clouds?** [1]

- ○ evaporation
- ● **condensation**
- ○ melting
- ○ freezing

3 **What drives the water cycle?** [1]

- ○ the Moon
- ○ the wind
- ● **energy from the Sun**
- ○ the sea

4 **Draw your own water cycle diagram. Label four processes and show the direction with arrows.** [4]

My water cycle

Sea → evaporation (arrows up) → condensation (clouds) → precipitation (rain) → collection (river to sea); transpiration from plants.

5 **Challenge. Explain why the water cycle is called *a cycle* and why the total amount of water on Earth stays the same.** [2]

The water goes round and round (evaporate → condense → fall → collect → evaporate); it only changes state and place, so none is made or destroyed.